

Solution Requirements

Date	04 October 2026
Team ID	POCKETSMART-TEAM
Project Name	PocketSmart AI – Smart Budget & Recommendation Assistant using Gemini Models
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	User Registration	Allow a user to register with username, email, optional full name and password; reject duplicate usernames and emails.	High
2	Authentication	Sign in with username and password and receive a JWT, also stored in an httpOnly cookie.	High
3	Session Management	Track the active session, allow sign-out that invalidates the token, and expire idle sessions.	High
4	Home Interior Planner	Accept total budget, counts of lights, fans, furniture and dining tables, rooms and extra notes; return a category-wise breakdown.	High
5	Party Planner	Accept budget, guests, party type, venue type and optional catering, decoration and entertainment needs; return a breakdown.	High
6	Jewelry Planner	Accept budget, occasion, preferences and an optional outfit image; return matching jewelry suggestions and styling tips.	High
7	Shopping Links	Attach search links for relevant Indian platforms to each recommended item.	Medium
8	Fallback Estimate	Return a rule-based estimate, flagged as a fallback, if Gemini fails.	High
9	Recommendation History	Save every result per user and allow listing and reopening of past plans.	Medium
10	Dashboard	Show the three planners and recent activity after sign-in.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Acceptance Criteria
1	Usability	Simple forms, INR formatting and a visible loading state.	A user can go from sign-in to a plan in a few steps.
2	Reliability	Try several Gemini models, retry once per model and fall back to a static estimate.	A usable result or a clear error is always returned.

3	Security	Hash passwords, keep the API key and secret in environment variables, restrict uploads.	Passwords are bcrypt hashes; only .jpg, .jpeg, .png and .webp up to 5 MB are accepted.
4	Data Integrity	Validate numeric and text inputs with Pydantic.	Zero or negative budgets and invalid guest counts are rejected.
5	Maintainability	Separate routes, authentication, AI logic and schemas.	Each concern lives in its own module.